Ziming Liu

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WORK EXPERIENCE

MetaCircle, China **May 2026 – Present**
- Founder & Chief Scientist

Shanghai Qizhi Institute, China **April 2026 – Present**
- Principal Investigator

Tsinghua University, China **April 2026 – Present**
- Assistant Professor at CollegeAI

Stanford University, USA **Sep 2025 – Feb 2026**
- Postdoc on Neuroscience + AI

EDUCATION

Massachusetts Institute of Technology, USA **Feb 2021 – August 2025**
- Ph.D. in Physics, working on Physics + AI

Microsoft Research Asia, China **Sep 2020 – Feb 2021**
- Research assistant in Machine Learning Theory Group

Peking University, China **Sep 2016 – June 2020**
- B.S. in physics, undergraduate research in high-energy physics

RESEARCH

Citations: 10,615 h-index: 25 i10-index: 38 ***(as of July 11, 2026)***

My research focuses on the foundations of artificial intelligence and next-generation AI systems. I pursue original research across four interconnected directions—Science of AI, AI for Science, and AI for AI—with the goal of uncovering the first principles of intelligence and advancing AI from an empirical discipline toward a scientific one.

On AI models and learning algorithms, I have developed several original architectures and frameworks, including Poisson Flow Generative Models (PFGM/PFGM++), Kolmogorov-Arnold Networks (KAN), and Brain-Inspired Modular Training (BIMT). Among them, KAN has become one of the most influential neural network architectures introduced in recent years, receiving over 4,000 Google Scholar citations and 15,000 GitHub stars. The PFGM series introduced a new paradigm for generative modeling based on Poisson fields, while BIMT explores biologically inspired modular training mechanisms to improve generalization and compositionality.

My work also investigates the theoretical foundations of deep learning, including scaling laws, grokking, learning dynamics, optimization, and interpretability, with the aim of identifying universal principles governing neural network training. I proposed the AI Poincaré framework for discovering physical laws, conservation laws, and symmetries using AI, and developed methods such as GenPhys to integrate physical inductive biases into generative and world models, bridging fundamental AI theory with AI for

Science. My research has been published at leading conferences including ICML, NeurIPS, and ICLR, has accumulated over 10,000 Google Scholar citations, and has been featured by MIT News, Quanta Magazine, Scientific American, Forbes, and New Scientist.

More recently, I have been pioneering the emerging direction of AI for AI, which aims to build AI systems that understand, predict, and optimize AI itself. To advance this vision, I founded Meta Circle, an AI research company developing next-generation infrastructure for AI research, including foundation models for AI, autonomous research systems, and open-source platforms that enable self-improving AI and accelerate scientific discovery.

AWARDS

- Google PhD fellowship, 2024-2026
- MIT graduate fellowship, 2021-2022
- Outstanding Graduate of Peking University, 2020

MEDIA COVERAGE

Kolmogorov-Arnold Networks

- Scientific American, "An Alternative to Conventional Neural Networks Could Help Reveal What AI Is Doing behind the Scenes"
- Quanta Magazine, "Novel Architecture Makes Neural Networks More Understandable"
- Github: most starred repo in May 2024 (15.6k stars)

Poisson Flow Generative Models

- Quanta Magazine, "The Physical Process That Powers a New Type of Generative AI"
- MIT News, "From physics to generative AI: An AI model for advanced pattern generation"

Understanding grokking

- Quanta Magazine, "How Do Machines 'Grok' Data?"

Learning symmetries

- New Scientist, "AI solves complex physics problems by looking for signs of symmetry"
- Nature Review Physics, "How machines could teach physicists new scientific concepts".

SERVICE

Reviewing

- Journals: Nature, Physical Review, Entropy, IEEE, Chaos
- Conferences: NeurIPS, ICLR, ICML

Organizing/Teaching

- AI4Science workshop at NeurIPS 2021, ICML 2022, and NeurIPS 2023.
- AI journal club at Physics of Living Systems (PLS) at MIT
- AI + Science journal club at the College of Computing at MIT
- (Tentative) Co-design and co-teach the "Physics of Deep Learning" course at MIT

Funding proposal

- Co-write NSF grant PHY-2019786 (IAIFI)

Mentoring undergraduate students

- Subhash Kantamneni (PhD @ Berkeley), Isaac Liao (PhD @ CMU), Ziqian Zhong (PhD @ CMU), Mingyang Deng (PhD @ MIT), Quan Nguyen (PhD @ Stanford), Vedang Lad (Researcher @ Stanford), Eric Gan

(Researcher @ OpenAI), Xinghong Fu, Riya, Tyagi, Xiaoman Ding, Carl Guo, Anish Mudide, Chloe Loughridge, Tara Kheirkhah, Mateja Vukelic, Rohan Mehta, Leo Yao.

SELECTED PAPERS

- Superposition Yields Robust Neural Scaling. **NeurIPS 2025 Best Paper Runner-up**
  Yizhou Liu, Ziming Liu, Jeff Gore
- **Kolmogorov-Arnold Networks Meet Science. Physical Review X (Featured in Physics)**
  Ziming Liu, Pingchuan Ma, Yixuan Wang, Wojciech Matusik, Max Tegmark
- **KAN: Kolmogorov-Arnold Networks. ICLR 2025 (Oral)**
  Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, Max Tegmark
- **Machine learning conservation laws from trajectories, Physical Review Letter (Editor's Suggestion), 2021**
  Ziming Liu, Max Tegmark
- **Towards understanding grokking: An effective theory of representation learning, NeurIPS (Oral), 2022**
  Ziming Liu, Ouail Kitouni, Niklas S Nolte, Eric Michaud, Max Tegmark, Mike Williams
- **Poisson flow generative models, NeurIPS, 2022**
  Yilun Xu*, Ziming Liu*, Max Tegmark, Tommi Jaakkola
- **The quantization model of neural scaling, NeurIPS 2022**
  Eric Michaud, Ziming Liu, Girit Uzay, Max Tegmark
- **Machine Learning Hidden Symmetries, Physical Review Letter (Editor's Suggestion), 2022**
  Ziming Liu, Max Tegmark
- **Seeing is Believing: Brain-Inspired Modular Training for Mechanistic Interpretability, Entropy, 2023**
  Ziming Liu, Eric Gan, Max Tegmark

SELECTED TALKS

- **"Unification of AI and Science: Towards AI that gets physics (fast)"**, Singapore AI tea talk (2025)
- **"KAN: Kolmogorov-Arnold Networks"**, Imagination In Action (2024), IAIFI summer workshop (2024)
- **"From Physical Processes to Generative Models"**, GenAI in STEM (2024)
- **"Discovering Physical Laws from Data"**, CMU ML seminar (2023)

FULL PUBLICATION LIST

In summary, the three pillars of my research are:

- **Science of AI**: Understanding AI using science, e.g., demystifying grokking [NeurIPS 2022, ICLR 2023]; neural scaling laws [NeurIPS 2022]; compositionality of generative models [NeurIPS 2023]; how do AI models do math [NeurIPS 2023], physics [Entropy 2024, Preprint 2025]; analogy between learning and physics [PRE 2025, Preprint 2025, NeurIPS 2024].
- **Science for AI**: Advancing AI using science, e.g., Kolmogorov-Arnold networks [ICLR 2024A, ICLR 2024B, Preprint 2024], Physics-inspired generative models [NeurIPS 2022, ICML 2023, preprint 2024], neuroscience-inspired modular networks [Entropy 2023, NeurIPS 2023], physics-inspired sampler/eigen-solver/optimizer [Preprint 2019, CMS 2024, PRE 2021, Preprint 2025].
- **AI for Science**: Advancing science using AI. Methods include discovering conservation laws [PRL 2021, PRE 2021, PRE 2022, PRE 2023, PRE 2024], hidden symmetries [PRL 2022], physics-augmented learning [PAL], and high-precision SciML [Entropy 2023]. Applications include math [ICLR 2024A], condensed matter physics [ICLR 2024A], periodic orbits [NeurIPS 2024], chemistry [PRE 2023], fluid mechanics [PRE 2023], photonics [SPIE], neuroscience [NeurIPS 2023], high-energy physics [PRR 2021], and program synthesis [Entropy 2023].

Review papers: AI in science [Nature 2023] and in physics [Review 2025].

Science of AI: Using scientific approaches to understand artificial intelligence

- **Physics of Skill Learning, arXiv: 2501.12391, 2025**
  Ziming Liu, Yizhou Liu, Eric J. Michaud, Jeff Gore, Max Tegmark
- **Fokker-Planck to Callan-Symanzik: evolution of weight matrices under training, arXiv: 2501.09659, 2025**
  Bu Wei, Kol Uri, Ziming Liu
- **How do Transformers Model Physics? Investigating the Simple Harmonic Oscillator, Entropy 2024**
  Subhash Kantamneni, Ziming Liu, Max Tegmark
- **On the expressiveness and spectral bias of KANs, ICLR 2025**
  Yixuan Wang, Jonathan Siegel, Ziming Liu, Thomas Hou
- **A Resource Model For Neural Scaling Law, ICLR 2024 BGPT workshop**
  Jinyeop Song*, Ziming Liu*, Max Tegmark, Jeff Gore
- **Do Diffusion Models Learn Semantically Meaningful and Efficient Representations?, NeurIPS 2023**
  Catherine Liang, Ziming Liu, Michell Ostrow, Ila R. Fiete
- **GenEFT: Understanding Statics and Dynamics of Model Generalization via Effective Theory, PRE 2025**
  David Baek, Ziming Liu, Max Tegmark
- **Grokking as Compression: A Nonlinear Complexity Perspective, NeurIPS 2023 UniReps workshop**
  Ziming Liu, Ziqian Zhong, Max Tegmark
- **A Neural Scaling Law from Lottery Ticket Ensembling, arXiv: 2310.02258, 2023**
  Ziming Liu, Max Tegmark
- **The Clock and the Pizza: Two Stories in Mechanistic Explanation of Neural Networks, NeurIPS (Oral), 2023**
  Ziqian Zhong*, Ziming Liu*, Max Tegmark, Jacob Andreas
- **The quantization model of neural scaling, NeurIPS, 2023**
  Eric J Michaud, Ziming Liu, Uzay Girit, Max Tegmark
- **Towards understanding grokking: An effective theory of representation learning, NeurIPS (Oral), 2022**
  Ziming Liu, Ouail Kitouni, Niklas S Nolte, Eric Michaud, Max Tegmark, Mike Williams
- **Omnigrok: Grokking beyond algorithmic data, ICLR (Spotlight), 2022**
  Ziming Liu, Eric J Michaud, Max Tegmark

Science for AI: Improving AI models (efficiency and interpretability) with science

- **Harmonic Loss Trains Interpretable AI Models**
  David Baek*, Ziming Liu*, Riya Tyagi, Max Tegmark
- **FOCUS: First Order Concentrated Updating Scheme, arXiv: 2501.12391, 2025**
  Yizhou Liu, Ziming Liu, Jeff Gore
- **KAN: Kolmogorov-Arnold Networks, ICLR 2025 (oral)**
  Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, Max Tegmark
- **Restart Sampling for Improving Generative Processes, NeurIPS, 2023**
  Yilun Xu, Mingyang Deng, Xiang Cheng, Yonglong Tian, Ziming Liu, Tommi Jaakkola
- **Seeing is Believing: Brain-Inspired Modular Training for Mechanistic Interpretability, Entropy**
  Ziming Liu, Eric Gan, Max Tegmark
- **GenPhys: From Physical Processes to Generative Models, arXiv: 2304.02637, 2023**
  Ziming Liu, Di Luo, Yilun Xu, Tommi Jaakkola, Max Tegmark
- **Pfgm++: Unlocking the potential of physics-inspired generative models, ICML, 2023**

Yilun Xu, Ziming Liu, Yonglong Tian, Shangyuan Tong, Max Tegmark, Tommi Jaakkola

- **Poisson flow generative models, NeurIPS, 2022**
  Yilun Xu*, Ziming Liu*, Max Tegmark, Tommi Jaakkola
- **Second order ensemble Langevin method for sampling and inverse problems, arXiv: 2208.04506, 2022**
  Ziming Liu, Andrew M Stuart, Yixuan Wang
- **Schrödinger principal-component analysis: On the duality between principal-component analysis and the Schrödinger equation, Physical Review E, 2021**
  Ziming Liu, Sitian Qian, Yixuan Wang, Yuxuan Yan, Tianyi Yang
- **Quantum-inspired hamiltonian monte carlo for bayesian sampling, arXiv: 1912.01937, 2019**
  Ziming Liu, Zheng Zhang

AI for Science: Accelerating scientific discoveries with AI

- **Do Two Scientists Agree?**
  Xinghong Fu, Ziming Liu, Max Tegmark
- **Interpretable Machine Learning in Physics: A Review**
  Sebastian Johann Wetzel, Seungwoong Ha, Iten Raban, Klopotek Miriam, Ziming Liu
- **KAN 2.0: Kolmogorov-Arnold Networks Meet Science. arXiv: 2408.10205**
  Ziming Liu, Pingchuan Ma, Yixuan Wang, Wojciech Matusik, Max Tegmark
- **Growing Brains: Co-emergence of Anatomical and Functional Modularity in Recurrent Neural Networks, NeurIPS 2023 UniReps workshop**
  Ziming Liu, Mikail Khona, Ila R. Fiete, Max Tegmark
- **Scientific discovery in the age of artificial intelligence, Nature, 2023**
  Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, …
- **Discovering New Interpretable Conservation Laws as Sparse Invariants, Physical Review E**
  Ziming Liu, Patrick Obin Sturm, Saketh Bharadwaj, Sam Silva, Max Tegmark
- **Precision machine learning, Entropy, 2023**
  Eric J Michaud, Ziming Liu, Max Tegmark
- **Machine learning conservation laws from differential equations, Physical Review E, 2022**
  Ziming Liu, Varun Madhavan, Max Tegmark
- **Machine learning hidden symmetries, Physical Review Letter (Editor's suggestion), 2022**
  Ziming Liu, Max Tegmark
- **Machine-learning nonconservative dynamics for new-physics detection, Physical Review E, 2021**
  Ziming Liu, Bohan Wang, Qi Meng, Wei Chen, Max Tegmark, Tie-Yan Liu
- **Physics-augmented learning: A new paradigm beyond physics-informed learning, NeurIPS 2021 AI4Science workshop, 2021**
  Ziming Liu, Yunyue Chen, Yuanqi Du, Max Tegmark
- **Applications of deep learning to relativistic hydrodynamics, Physical Review Research, 2021**
  Hengfeng Huang, Bowen Xiao, Ziming Liu, Zeming Wu, Yadong Mu, Huichao Song
- **Machine learning conservation laws from trajectories, Physical Review Letter (Editor's Suggestion), 2021**
  Ziming Liu, Max Tegmark
- **Robustness of principal component analysis of harmonic flow in heavy ion collisions, Physical Review C, 2020**
  Ziming Liu, Arabinda Behera, Huichao Song, Jiangyong Jia
- **Principal component analysis of collective flow in relativistic heavy-ion collisions, European Physical Journal C, 2019**
  Ziming Liu, Wenbin Zhao, Huichao Song